On the validation of the TDAC method for a stochastic turbulent combustion model in OpenFOAM

Zhijie Huo, Sebastian Galindo-Lopez Matthew J. Cleary, Assaad R. Masri

The University of Sydney, NSW 2006, Australia

THE UNIVERSITY OF SYDNEY

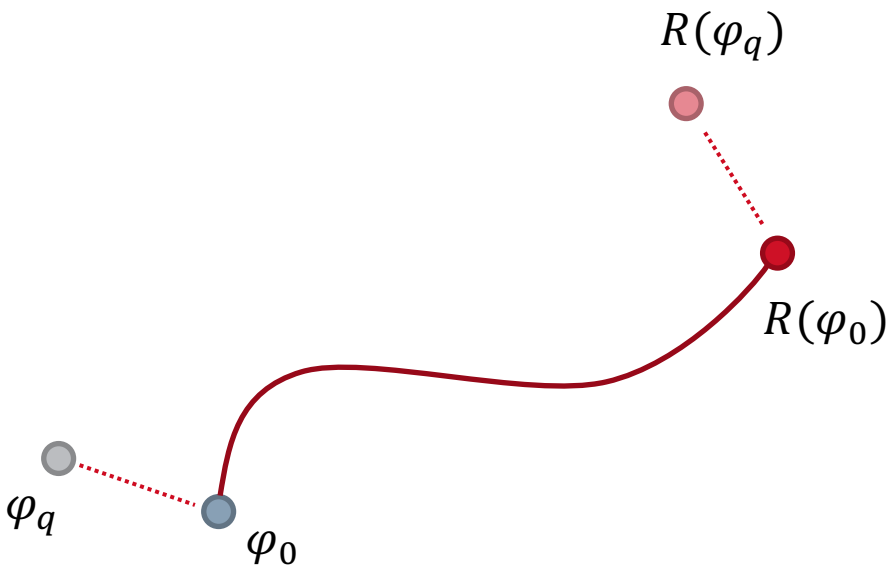
Tabulation Dynamic Adaptive Chemistry (TDAC)
Contino et al. (2011)

in situ Adaptive Tabulation (ISAT)
Pope (1997)

Dynamic Adaptive Chemistry (DAC)
Liang et al. (2009)

- TDAC is a combination of ISAT and DAC
  - ISAT stores computation results and approximate future similar calculations
  - DAC reduces the dimension of chemistry kinetics

in situ Adaptive Tabulation (ISAT)

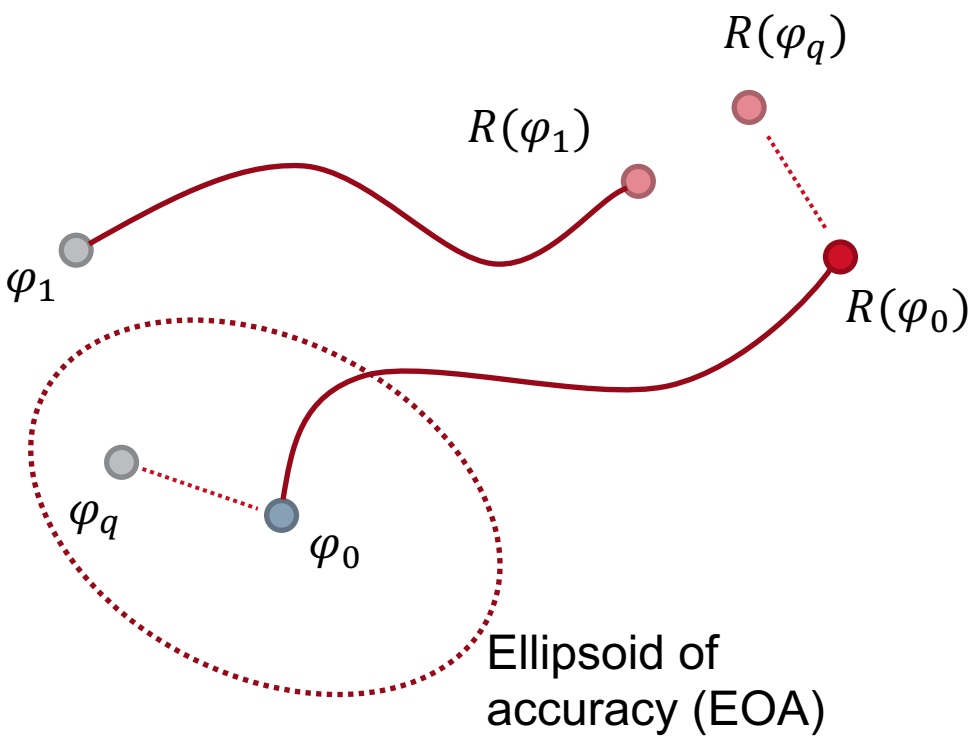


Reaction mapping $R(\varphi_0)$ is computed by direct integration (DI) from φ_0 and is stored in a binary tree such that one collection of these data is called a leaf. If a later query point φ_q is similar to φ_0, its reaction mapping is approximated by

$$R(\varphi_q) \approx R(\varphi_0) + A(\varphi_0 - \varphi_q)$$

Where A is a mapping gradient matrix: $A_{ij} = \partial R_i(\varphi^0)/\partial \varphi_j$.

in situ Adaptive Tabulation (ISAT)



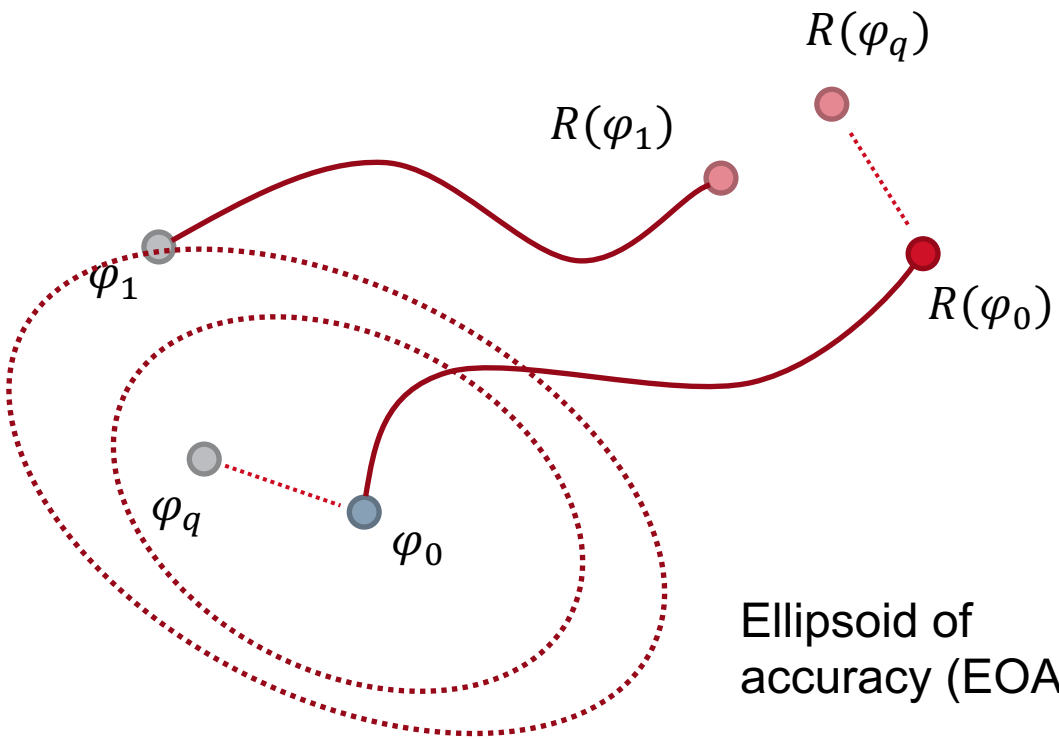
A hyper-ellipsoid centered at φ_0 is defined as ellipsoid of accuracy (EOA) within which the approximation of

$$R(\varphi_q) \approx R(\varphi_0) + A(\varphi_0 - \varphi_q)$$

is valid.

If no approximation is valid, such φ_1, DI is used to compute the reaction mapping $R(\varphi_1)$.

in situ Adaptive Tabulation (ISAT)

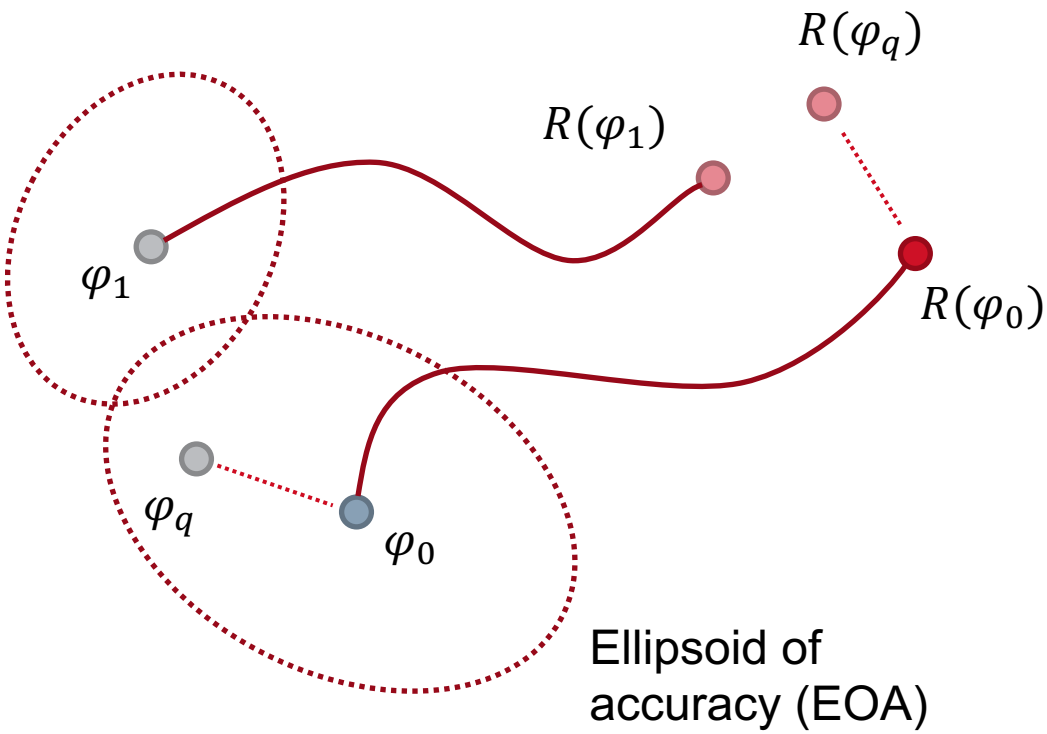


Ellipsoid of accuracy (EOA)

$R(\varphi_1)$ is then compared to $R(\varphi_0)$, If R(φ_0) - R(φ_1) < $tolerance$, the EOA is considered to be too conservative and a growth operation is executed to adjust the EOA.

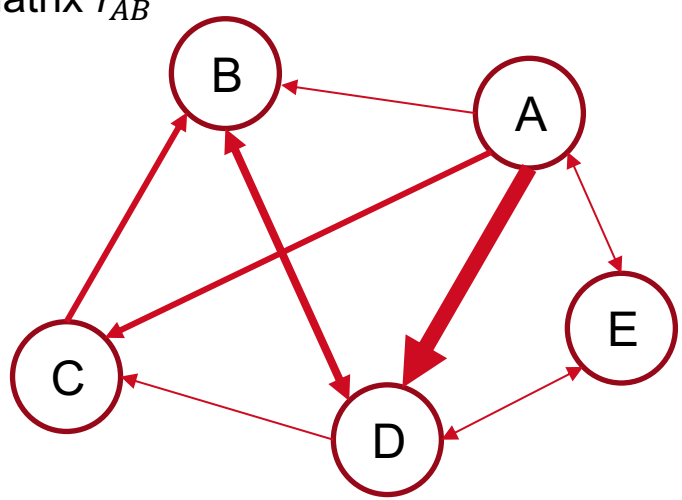
in situ Adaptive Tabulation (ISAT)

However, if
R(φ_0) - R(φ_1) > $tolerance$,
A new EOA is added for future approximation.



Dynamic Adaptive Chemistry (DAC)

The strength of the kinetic relationships between the species A and B is indicated schematically by the width of the edges in a direct relation graph (DRG) and is defined by a contribution matrix r_{AB}



$$r_{AB} = \frac{\left|\sum_{i=1}^{n_r} \nu_{i,A}\omega_i\delta_B^i\right|}{\max(P_A, C_A)}$$

$$P_A = \sum_{i=1}^{n_r} \max(0, \nu_{i,A}\omega_i)$$

$$C_A = \sum_{i=1}^{n_r} \max(0, -\nu_{i,A}\omega_i)$$

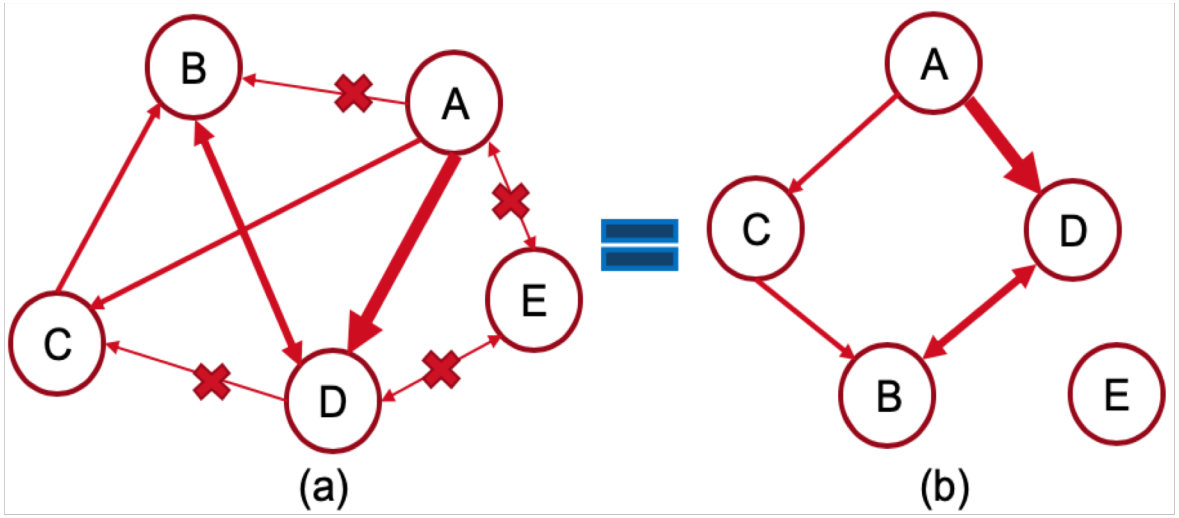
$\nu_{A,i}$ *is stoichiometric coefficient of species A in reaction i,*

ω_i *is the production rate,*

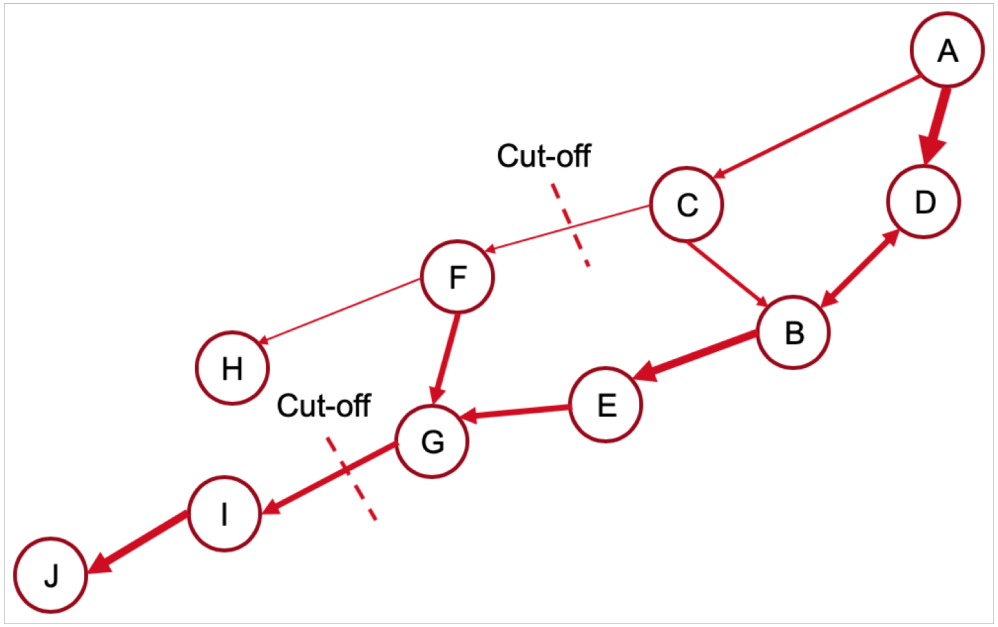
$$\delta_{Bi} = \begin{cases} 1 & \text{if the } i\text{th elementary reaction} \\ & \text{involves species } B, \\ 0 & \text{otherwise.} \end{cases}$$

Dynamic Adaptive Chemistry (DAC)

To reduce the dimension of the composition space edges for which $r_{AB} < \varepsilon_{DAC}$ are eliminated and the DRG can be reconfigured as shown in (b). Different group is kept depending on the user defined search initial set (SIS)



Dynamic Adaptive Chemistry (DAC)



To take advantage of the additional dimension reduction that is possible, error propagation is applied to further trim the DRG. The dependency of a secondary species P from a primary species P_0 is quantified by

$$R_{P_0}(P) = \max_{\Omega}\left(\prod r_{ij}\right)$$

where Ω is all possible paths from P_0 to P and r_{ij} is the contribution matrix, Species with $R_{P_0}(P) < \varepsilon_{DAC}$ are also marked inactive and eliminated.

Tutorial

- Tutorial case is in flamed/DRM19-TDAC
- TDAC related input parameters are all specified in constant/chemistryProperties and will only be activated when finiteRateParticleChemistry is used.
- ISAT and DAC can be turned on separately within tabulation and reduction section.
- Direct integration is used when both tabulation and reduction are switched off:

```
reduction
{
active off;
method pNone;
}
```

```
tabulation
{
active off;
method pNone;
}
```

Tutorial - Reduction scheme setup

```
active on; // Activate reduction

log on; // Switch logging of the reduction statistics and performance

tolerance 0.0001;

method pDAC; // Dynamic Adactive Chemistry (DAC)
// this is an important set of parameters for chemistry reduction.
// All species and reactions that strongly (depends on the tolerance) affect
// the production(or consumption) of these species will be included.
initialSet
{
CO ;        CH4 ;       HO2 ;
}
```

Tutorial - Reduction scheme setup

```
// For DAC, option to automatically change the
// Search initiating set (SIS) switch from HO2 to H2O
// and CO to CO2, + disable fuel
automaticSIS off;

// When automaticSIS, the method needs to know the fuel
//by specifying species and mass fraction
fuelSpecies
{
CH4        1;
}
```

Tutorial - Tabulation scheme setup

```
// Activate tabulation
active on;

// Switch logging of the tabulation statistics and performance
log on;

printProportion off;

printNumRetrieve off;

// Tolerance used for retrieve and grow
tolerance 1e-4;

// ISAT is the only method currently available
method pISAT;
```

Tutorial - Tabulation scheme setup

```
// Scale factors used in the definition of the ellipsoid of accuracy
scaleFactor
{
otherSpecies 1;
Temperature 2400;
Pressure 1e6;
deltaT 1;
}
// Maximum number of leafs stored in the binary tree
maxNLeafs 200000;

//leaves will be deleted if they have not been used by this number of timestep.
chPMaxLifeTime 20;

// Maximum number of growth allowed on a leaf
maxGrowth 20000;
```

Tutorial - Tabulation scheme setup

```
// Number of time steps between analysis of the tree to remove old
// chemPoints or try to balance it
checkEntireTreeInterval 500;

// Parameters used to decide whether to balance or not if the tree's depth
// is larger than maxDepthFactor*log2(nLeafs) then balance the tree
maxDepthFactor 2;

// Try to balance the tree only if the size of the tree is greater
minBalanceThreshold 30;

// Activate the use of a MRU (most recently used) list
MRURetrieve false;

// Maximum size of the MRU list
maxMRUSize 0;
```

Tutorial - Tabulation scheme setup

```
// Allow to grow points?
growPoints true;

// When mechanism reduction is used, new dimensions might be added
// maxNumNewDim set the maximum number of new dimensions added during a growth
maxNumNewDim 10;
// Secondary search in binary tree
max2ndSearch 0;

// Clean the binary tree when it is full.
// if MRU is not activatied, cleaning will delete the whole binary tree
cleanAfterFull false;

// Used to do tests. Direct integration will be done after retrieve
// Results are comparied and ensamble results are stored in "./TDAC/Summary"
TestSwitch false;//true;
```